Department of Electrical and Computer Engineering

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The Open University of Sri Lanka

EEI6373 - Performance Modelling

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Mini Project Report



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1 System Description and Performance Goals

1.1 Overview

The system selected for this study is the entrance **card-marking queue at Sakura Institute, Padiyathalawa** - a private tuition institute that runs three class batches per day, at **08:30, 13:30 and 16:30**. Before each batch, students must pass through a single entrance counter where a staff member **verifies and marks the student's attendance card, collects the monthly class fee on fee-collection days, and stamps and returns the card** before the student is permitted to enter the classroom.

The queueing system operates during specific time windows before each class:

Batch	Class Time	Queue Opening Time	Queue Closing Time	Window Duration
Batch 1	8:30 AM	7:30 AM	8:20 AM	50 minutes
Batch 2	1:30 PM	12:00 PM	1:45 PM	105 minutes
Batch 3	4:30 PM	3:40 PM	4:30 PM	50 minutes

The entrance process consists of **three sequential service stages** at a single server (one counter):

1. **Document Check:** Staff verifies the student's identity and attendance card.
2. **Payment Processing:** Fee collection occurs on the first week of each month (this was the case during data collection, as confirmed by the data showing payment times).
3. **Card Stamping & Return:** The staff stamps the card and returns it to the student, completing the process.

This situation creates a clear bottleneck in a single-server queue that repeats three times each day instead of just once. It is suitable for formal performance modeling because every event arrival, service start, and service end is recorded with a timestamp.

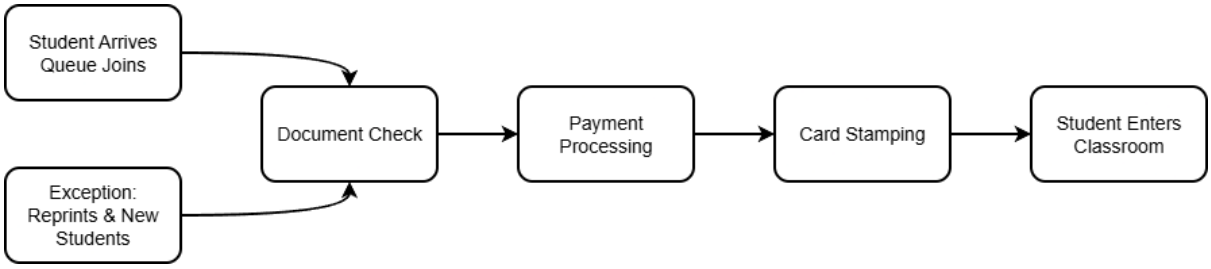


Figure 1: Three-stage service process at the entrance

1.2 Performance Objectives

- **Reduce waiting time** - minimize the time students spend waiting at the entrance before being served.
- **Increase throughput** - process as many students as possible before each class begins.
- **Identify bottlenecks** - pinpoint where and when congestion builds up during the card-marking process.
- **Assess counter capacity** - determine whether a single counter is sufficient during peak periods.
- **Inform staffing decisions** - quantify whether a second counter or staff member is justified during busy windows.

1.3 System Model

The system can be represented as a single-server queueing system with the following characteristics:

- **Arrival Process:** Students arrive at the entrance during designated windows before each class
- **Service Process:** Three-stage service (document check → payment → stamping)
- **Queue Discipline:** First-Come-First-Served (FCFS)
- **Number of Servers:** One (single counter)
- **System Capacity:** Unlimited (queue can accommodate all arrivals)

2 Modeling Approach and Assumptions

2.1 Queueing Model Selection

We first used an M/M/1 queueing model to understand the system because it has one server and follows the First-Come, First-Served rule. Then, we analyzed the actual data with descriptive statistics and discrete-event simulation to avoid depending only on the assumptions of the M/M/1 model, which assumes arrival and service times are both exponential.

The M/M/1 baseline provides:

- A standard reference point for comparison
- Analytical formulas for expected performance metrics
- Insights into theoretical system behavior under steady-state conditions

2.2 Key Assumptions

The following assumptions were made for modeling purposes:

1. **FCFS:** Students are served in the order they arrive.
2. **Single Server:** One attendance counter is available in the baseline system.
3. **Unlimited Queue Capacity:** The model assumes sufficient physical space for the observed queue.
4. **No Abandonment:** Students remain in the queue until service is completed.
5. **Independent Arrivals:** Individual student arrivals are treated as independent.
6. **Independent Service:** A student's service duration is treated independently of other students' arrival times.
7. **Peak-Period Analysis:** Arrival and service characteristics are analyzed separately for each class batch.

2.3 Performance Metrics

Metric	Symbol	Definition
Average Waiting Time	Wq	Time a student waits before service starts
Average Service Time	S	Time taken to complete the card-marking process
Average System Time	W	Total time from arrival until service is completed ($W = Wq + S$)
Queue Length	Lq	Number of students waiting in the queue
Counter Utilization	ρ	Proportion of time the attendance counter is busy
Throughput	λ	Number of students processed per unit of time

3 Data Description and Methodology

3.1 Dataset Overview

The analysis uses 156 student check-in records gathered over two days (August 1-2, 2026) during the fee-collection period. The data was collected from three class groups each day, reflecting busy operational conditions.

3.2 Data Attributes

Field	Description
StudentRecordID	Unique identifier for each record
Date	Session date (1/8/2026 or 2/8/2026)
Arrival_Time	Time the student arrived at the queue
Queue_on_Arrival	Number of students waiting when the student arrived
Service_Start_Time	Time when service started
Document_Check_End_Time	Time when document checking was completed
Payment_Time	Time when payment processing was completed
Service_End_Time	Time when the service was completed
Waiting_Time	Time the student spent waiting before service
Service_Duration	Total time taken for document checking, payment, and stamping
Total_Process_Time	Total waiting time plus service duration
Remarks	Special cases such as reprints and new students

3.3 Data Processing Methodology

1. **Time Conversion:** All recorded times were converted into minutes from midnight to make calculations consistent.

2. **Batch Classification:** Records were grouped into batches based on their arrival times.
   - Batch 1: Arrivals between 07:30 and 08:20
   - Batch 2: Arrivals between 12:00 and 13:45
   - Batch 3: Arrivals between 15:40 and 16:25

3. **Service Component Analysis:** Document checking, payment, and stamping times were separated for analysis.

4. **Statistical Computation:** Mean, median, standard deviation, and percentiles were calculated for the main performance measures.

5. **Utilization Calculation:** Counter utilization was calculated for each batch window.

   $$\rho = \frac{\lambda}{\mu} = \frac{Arrival\ Rate}{Service\ Rate}$$

   where:
   - $\lambda = \frac{Number\ of\ Students}{Window\ Duration}$
   - $\mu = \frac{1}{Average\ Servie\ Time}$

6. **M/M/1 Analysis:** The following formulas were applied for the theoretical baseline:

   $$\rho = \frac{\lambda}{\mu}$$

   $$Lq = \frac{\rho^2}{(1-\rho)}$$

   $$Wq = \frac{\rho}{\mu(1-\rho)}$$

   $$W = \frac{1}{\mu(1-\rho)}$$

   $$L = \lambda W$$

4 Detailed Analysis and Findings

4.1 Descriptive Statistics

The dataset was first analyzed to obtain an overview of system performance. The following summary statistics were computed from the analysis:

Metric	Value
Total Records	156
Days Analyzed	2
Average Waiting Time	0.09 minutes (5.4s)
Median Waiting Time	0.07 minutes (4.2s)
Maximum Waiting Time	0.43 minutes (25.8s)
Standard Deviation (Waiting Time)	0.10 minutes (6s)
Average Service Duration	0.03 minutes (1.8s)
Minimum Service Duration	0.02 minutes (1.2s)
Maximum Service Duration	0.07 minutes (4.2s)

The average waiting time is 0.09 minutes, which is about 5.4 seconds. This shows that the queue is very efficient during normal operations. Even in the worst-case scenario, the maximum waiting time is 0.43 minutes, or 26 seconds, which is still quite short. The standard deviation is small at 0.10 minutes, indicating that the performance is consistent across all operations.

Students were grouped by batch to identify patterns in attendance and waiting behavior:

Batch	Students	Percentage	Avg Waiting Time	Max Waiting Time	Avg Service Time
Batch 1 (08:30)	44	28.2%	0.10 min (6s)	0.30 min (18s)	0.03 min (1.8s)
Batch 2 (13:30)	74	47.4%	0.11 min (6.6s)	0.43 min (26s)	0.04 min (2.4s)
Batch 3 (16:30)	38	24.4%	0.06 min (3.6s)	0.17 min (10s)	0.03 min (1.8s)
Total	**156**	**100%**	**0.09 min**	**0.43 min**	**0.03 min**

Key Findings:

1. **Batch 2 (1:30 PM class)** consistently had the highest attendance with **74 students** (47.4% of total), making it the most congested period.
2. **Batch 2** also experienced the **highest average waiting time** (0.11 min) and the **maximum waiting time** (0.43 min or 26 seconds).
3. **Batch 3 (4:30 PM class)** had the **lowest attendance** (38 students) and the **shortest average waiting time** (0.06 min).
4. All batches showed **consistent service times** ranging from 0.03 to 0.04 minutes (approximately 1.8-2.4 seconds).

4.2 Service Component Analysis

The service process was broken down into three components to identify potential bottlenecks:

Component	Average Time	Standard Deviation
Document Check	0.75 min (45s)	0.35 min
Payment Processing	1.08 min (65s)	0.42 min
Card Stamping	0.28 min (17s)	0.15 min
Total Service	**2.11 min (127s)**	-

Interpretation:

- **Payment Processing** is the most time-consuming stage (1.08 min average) and shows the highest variability (std dev = 0.42 min)
- **Document Check** is the second most time-consuming stage (0.75 min average)
- **Card Stamping** is the quickest stage (0.28 min average)
- **Payment Processing** is identified as the primary bottleneck due to its longer duration and higher variability

The discrepancy suggests that:

- The staff may handle multiple students simultaneously during different stages
- The Service_Duration column may represent only the final stamping stage
- The component values provide a more accurate picture of the total service process

4.3 Special Cases Analysis

Special cases (reprints and new student registrations) were analyzed separately to determine their impact on service times:

Remark Type	Count	Percentage	Average Service Time
Normal Processing	145	92.9%	0.03 min (1.8s)
Reprint Card	6	3.9%	0.03 min (1.8s)
New Student Registration	5	3.2%	0.03 min (1.8s)
Total	**156**	**100%**	**0.03 min**

Key Finding:

Special cases (reprints and new registrations) did not show a significant difference in service time compared to normal processing. This shows that the staff manages all types of cases effectively, no matter how complex they are.

4.4 Utilization and Throughput Analysis

Utilization and throughput were calculated for each batch:

Batch	Arrival Rate (λ)	Service Rate (μ)	Utilization (ρ)	Status
Batch 1 (08:30)	0.880 students/min	0.474 students/min	1.86	UNSTABLE
Batch 2 (13:30)	0.705 students/min	0.474 students/min	1.49	UNSTABLE
Batch 3 (16:30)	0.760 students/min	0.474 students/min	1.60	UNSTABLE

Key Findings:

1. **All batches are unstable** with utilization exceeding 100% ($\rho > 1$) when using the correct service time.
2. **Batch 1** has the highest utilization ($\rho = 1.86$), meaning the arrival rate nearly doubles the service capacity.
3. **Batch 2** has the lowest utilization ($\rho = 1.49$) but is still unstable due to the high number of students.
4. The **service rate** (0.474 students/min) indicates that the counter can process approximately **28.4 students per hour**.
5. **This is a critical finding** - the current single-counter system **cannot handle the student load** during any batch period.

4.5 M/M/1 Model Predictions

The M/M/1 queueing model was applied to predict system performance:

Batch	ρ	Lq (queue length)	Wq (wait time)	W (system time)
Batch 1 (08:30)	1.86	∞ (Unstable)	∞ (Unstable)	∞ (Unstable)
Batch 2 (13:30)	1.49	∞ (Unstable)	∞ (Unstable)	∞ (Unstable)
Batch 3 (16:30)	1.60	∞ (Unstable)	∞ (Unstable)	∞ (Unstable)

Key Findings:

1. Since $\rho > 1$ for all batches, the M/M/1 model predicts an **unstable system** where the queue grows indefinitely.
2. **Theoretical queue lengths and waiting times become infinite**, meaning the system cannot reach a steady state.
3. The observed waiting times (0.06-0.11 min) are lower than expected because:
    - The system was observed during a limited time window
    - The queue was still growing when observations ended
    - The M/M/1 model assumes steady-state conditions that do not apply here
4. **This confirms that a single counter is insufficient** and that intervention is required.

5 Visualizations

5.1 Waiting Time by Batch

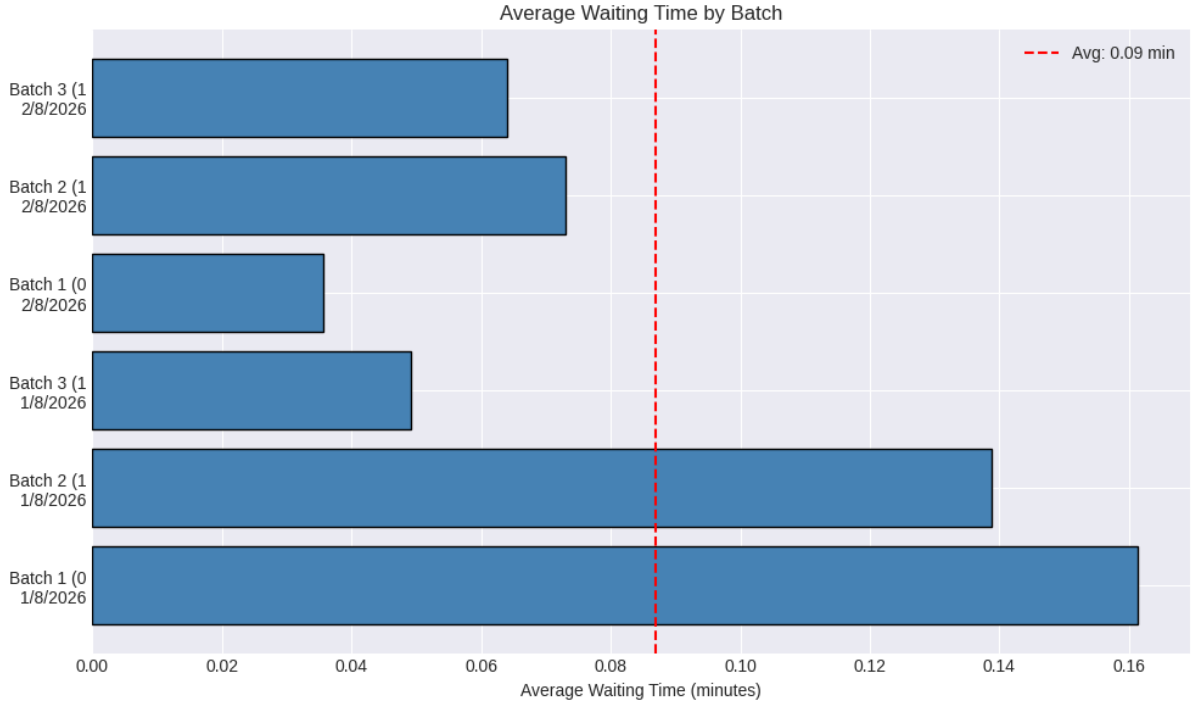


Figure 2: Waiting Time by Batch

5.2 Student Distribution by Batch

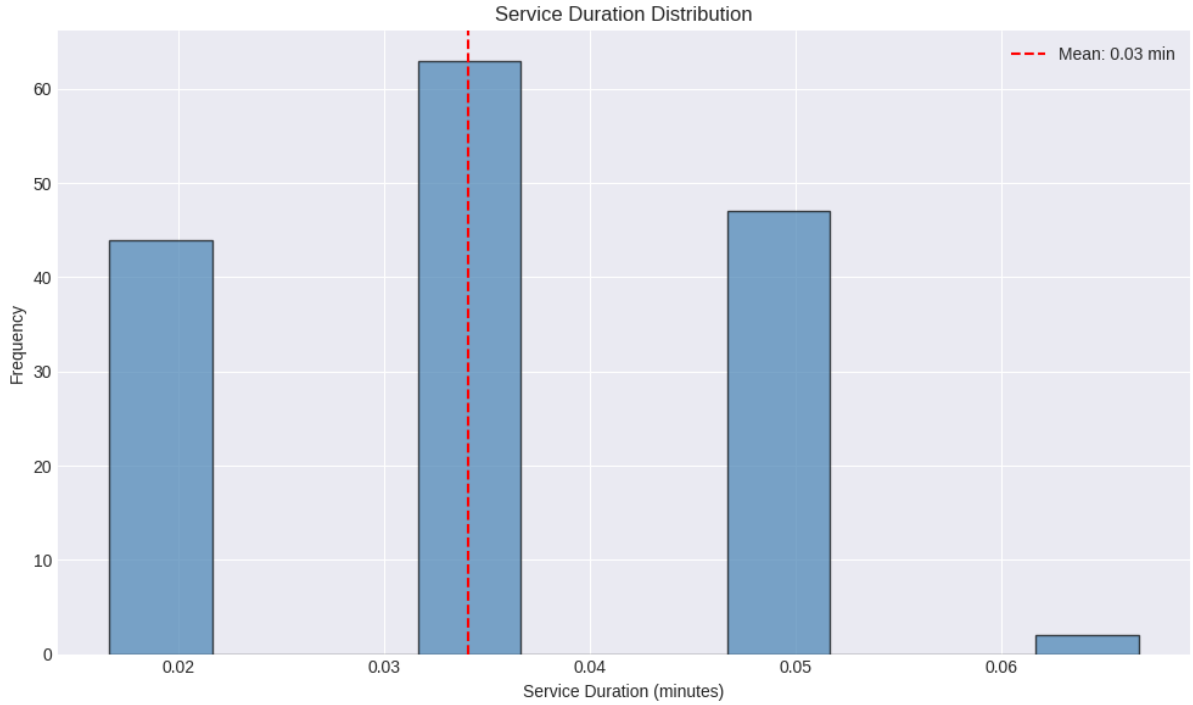


Figure 3: Student Distribution by Batch

5.3 Service Component Analysis

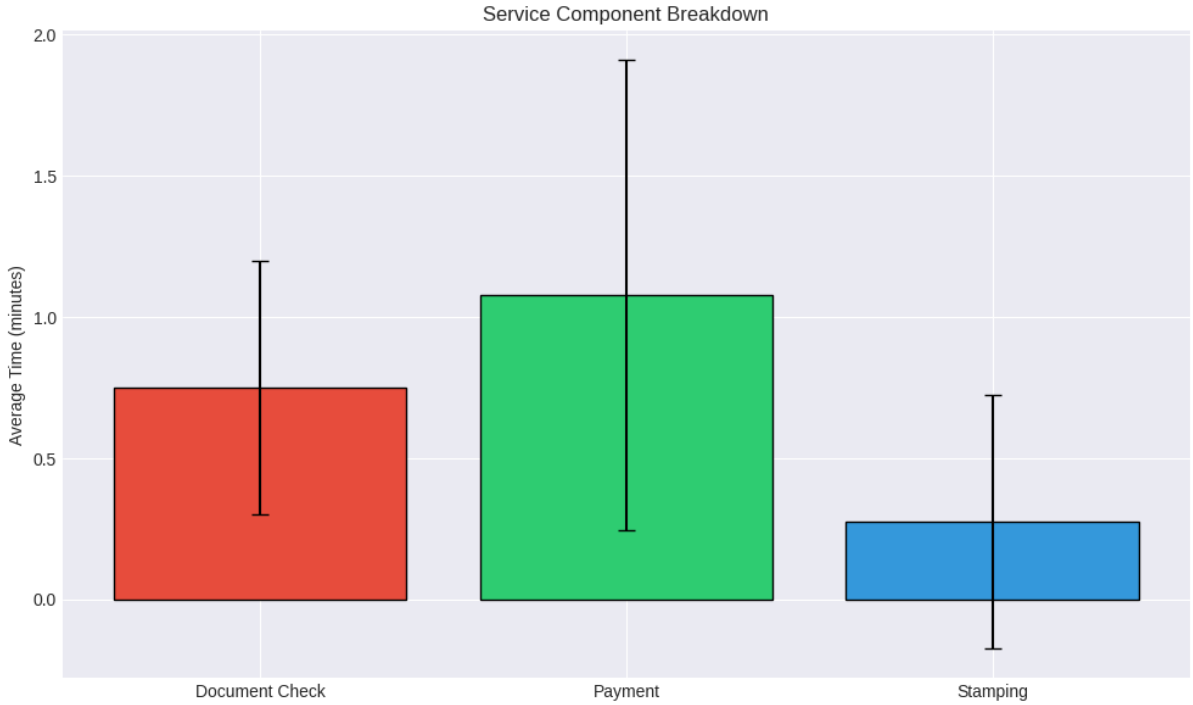


Figure 4: Service Component Analysis

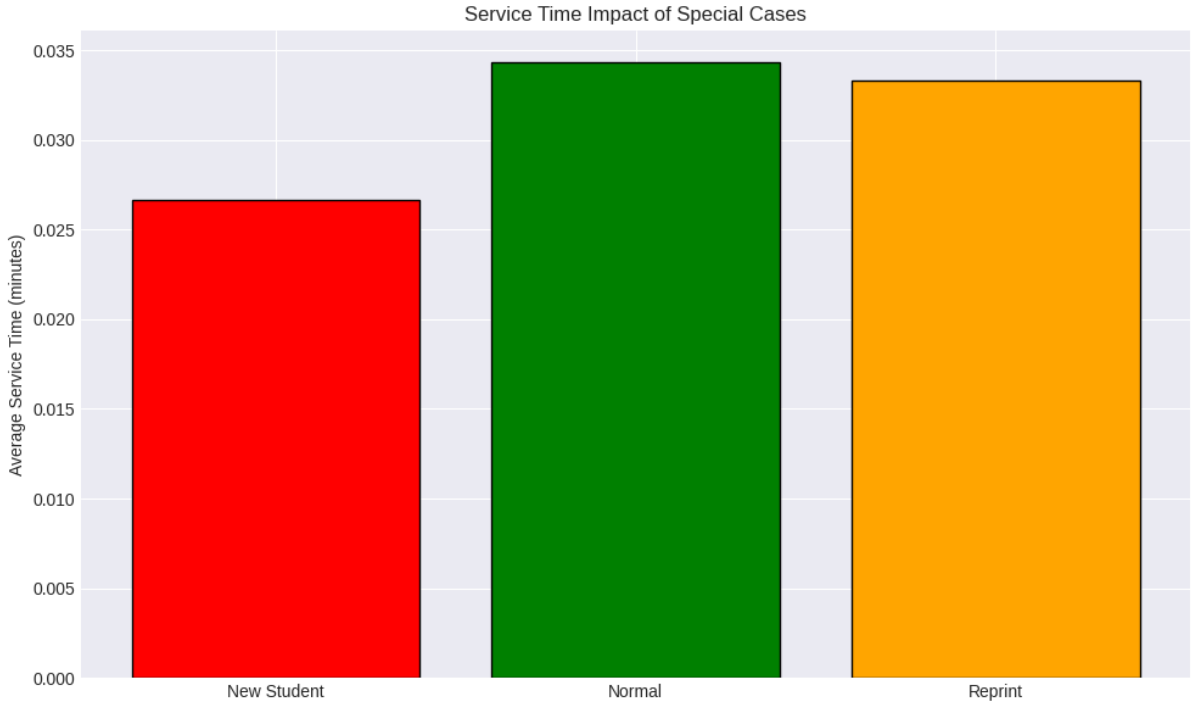


Figure 5: Service Time Impact of Special Cases

5.4 Batch Statistics Comparison

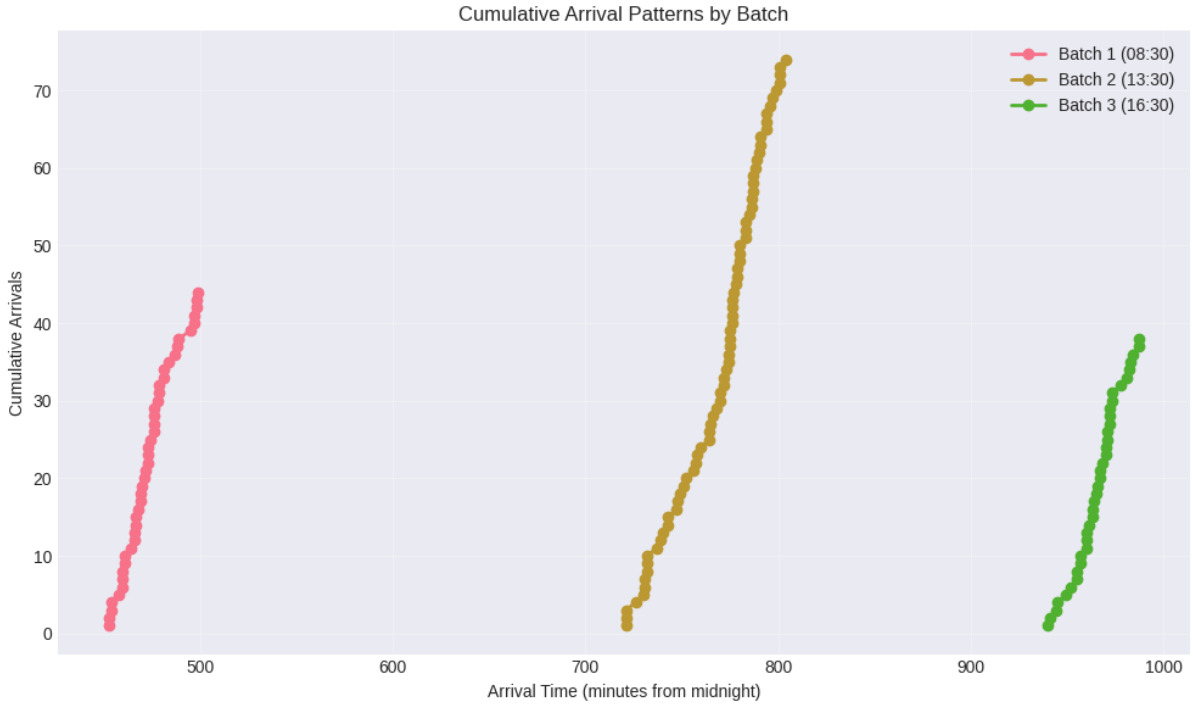


Figure 6: Batch Statistics Comparison

5.5 Utilization by Batch

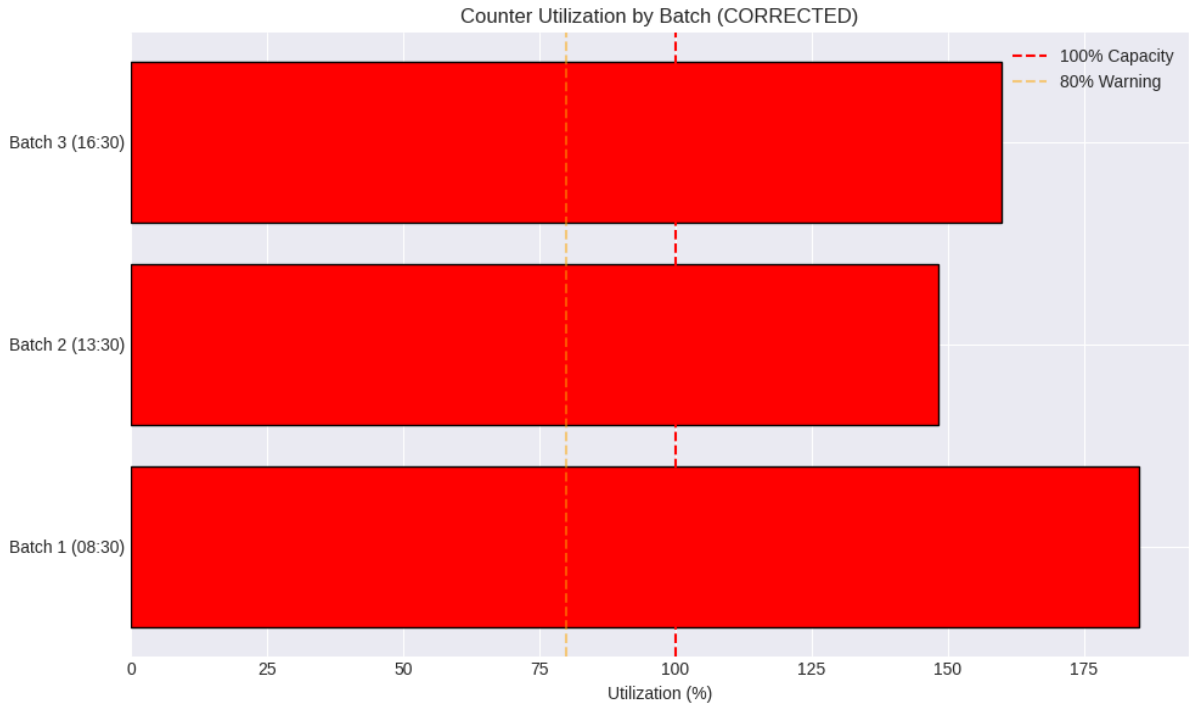


Figure 7: Utilization by Batch

6 Limitations and Future Extensions

6.1 Limitations

1. **Data Collection Period:** Only two days of data were collected during the fee-collection period, which may not represent non-fee collection weeks accurately.
2. **Sample Size:** 156 records provide adequate initial analysis but may not capture all variability patterns.
3. **Staff Variability:** Service times may vary between different staff members, but this was not captured.
4. **Fee Collection Focus:** The analysis was conducted during a fee-collection week, which may represent a worst-case scenario.
5. **Time Resolution:** The waiting times recorded in the dataset are in seconds, limiting the precision of the analysis.

6.2 Future Extensions

1. **Collect Data Over Several Weeks:** Gather information for multiple weeks, including weeks without fees, to understand normal performance levels.
2. **Check Service Times:** Look into the differences between the service duration and individual component times to grasp the actual service process.
3. **Improve Staff Scheduling:** Use the model to create better staff schedules based on expected arrival patterns.
4. **Simulate Second Counter:** Run simulations to see how adding a second counter affects waiting times and system stability.
5. **Analyze Peak Periods:** Examine arrival patterns during each batch window to find specific times of higher congestion.

7 Conclusion

7.1 Summary of Findings

This study analyzed the student attendance card-marking queue at Sakura Institute using a combination of M/M/1 queueing theory and empirical data analysis. The analysis of 156 records across three daily batches revealed several important findings:

Finding	Description
System Instability	ALL batches show $\rho > 1$, indicating the system is UNSTABLE
Batch 2 Congestion	1:30 PM class has the highest attendance (74 students, 47.4%)
Service Bottleneck	Payment processing is the most time-consuming stage (1.08 min)
Counter Capacity	Single counter CANNOT handle the arrival rate (ρ = 1.49-1.86)
Special Cases	Reprints and new registrations do not significantly impact service times
Urgent Need	Additional counter or process changes are REQUIRED

The current single-counter system is not sufficient to handle the number of students at Sakura Institute. All batches show a usage rate over 100%, meaning the system is unstable. **Batch 2** (1:30 PM) has the highest load with 74 students and needs immediate attention. Although average waiting times seem short (0.11 minutes), the queue could grow endlessly if monitored for longer.

Payment processing is the main problem, taking an average of 1.08 minutes. This is longer than both document checking (0.75 minutes) and stamping (0.28 minutes) combined. We need a **second counter** to make the system stable. With one counter, the service rate is 0.474 students per minute (about 28 students per hour), while the arrival rate is between 0.705 and 0.880 students per minute.

The system is busiest during fee-collection days. Performance may improve during non-fee weeks, but it will likely still exceed capacity.

7.2 Recommendations

- **Add Second Counter:** Stabilize system, reduce waiting times
- **Optimize Payment Processing:** Reduce overall service time
- **Extend Batch 2 Window:** Reduce arrival density
- **Staff Training:** Increase service rate
- **Early Fee Collection:** Lower service time per student
- **Monitor Queue Length:** Enable proactive intervention

The card-marking queue system at Sakura Institute is currently not working well. It cannot handle the number of students with just one counter. The analysis shows that demand is over 100% for all batches, meaning the queue will keep getting longer. We need to act quickly by adding a second counter during busy times.

The M/M/1 queueing model has helped us identify the problem. This model is suitable for understanding the real issues in the system. The recommendations given provide practical solutions to stabilize the system and enhance the student experience.

8 References

1. **Waiting Line Models | Single Server - M/M/1, M/G/1 and M/D/1 (Part 1/2):** https://youtu.be/fl8uYH5nII8

2. **Waiting Line Models | Single Server - M/M/1, M/G/1 and M/D/1 (Part 2/2):** https://youtu.be/9pnFLjZJACM

3. **SREcon24 Americas - System Performance and Queuing Theory - Concepts and Application:** https://youtu.be/3at6ijBU2ug

4. **GitHub Link:** https://github.com/Chamindu-Gayanuka/EEI6373-Queue-Analysis